

Climate Change: Causes, Effects, and Solutions

Climate change refers to long-term shifts in global temperatures and weather patterns. While natural factors play a role, scientific consensus confirms that human activities have been the dominant driver of climate change since the mid-20th century.

1. Greenhouse Gas Emissions

The burning of fossil fuels—coal, oil, and natural gas—releases carbon dioxide (CO₂) and methane (CH₄) into the atmosphere. These greenhouse gases trap heat, raising the Earth's average temperature. CO₂ concentrations have risen from 280 ppm before industrialisation to over 420 ppm today.

2. Observed Impacts

Consequences of rising temperatures include: rising sea levels due to glacial melt, more frequent and intense extreme weather events (hurricanes, droughts, heatwaves), ocean acidification harming marine ecosystems, and shifts in biodiversity as species migrate toward cooler habitats.

3. The Paris Agreement

Adopted in 2015, the Paris Agreement commits nations to limit global warming to well below 2 degrees Celsius above pre-industrial levels, with efforts to stay below 1.5 C. Countries submit Nationally Determined Contributions (NDCs) outlining their climate action plans, which are updated every five years.

4. Renewable Energy Transition

Solar and wind power have seen dramatic cost reductions—solar photovoltaic costs dropped by nearly 90 percent in the decade to 2023. Electric vehicles, green hydrogen, and energy storage technologies are also critical components of the low-carbon transition.

5. Carbon Capture and Removal

Beyond emissions reductions, negative-emission technologies such as direct air capture (DAC), bioenergy with carbon capture and storage (BECCS), and natural solutions like reforestation and wetland restoration are considered necessary to meet climate targets.